CostVision Rule Library

DFM / DFA rules, geometry advisors and idea-generation levers — extracted from the engine source on 06 Aug 2026

Every rule in this document was read out of the code that runs, by `scripts/extract-rule-library.ts`. Nothing is transcribed. Each entry shows the exact condition that fires it, the severity and category the engine assigns, the saving it claims, and the recommendation the engineer sees. Re-running the extractor after a rule change regenerates this document against the new code, so it cannot drift out of date the way a hand-written rule sheet does.

Layer	Count	Source file	What it does
DFM rules	41	src/engine/dfm-dfa.ts	Threshold checks on the finished estimate: utilisation, OEE, cycle, tooling, tolerance, packaging, logistics.
DFA rules	11	src/engine/dfm-dfa.ts	Assembly and handling checks in the Boothroyd tradition: part count, stations, setups, manning, fastening.
Geometry advisors	10 modules 75 checks	src/engine/modules/*-advisor.ts	Per-process physics: wall thickness, draft, section ratio, bend radius, undercuts, gate/runner, ply schedule.
Cost optimisations	11	src/engine/dfm-dfa.ts	Cost-structure levers raised alongside the DFM findings.
Idea levers	26	src/engine/idea-levers.ts	360-degree catalogue across material, design, process, tooling, logistics, commercial, quality, sustainability.
TOTAL	164		Deterministic checks run on every estimate. No AI involved in any of them.

How the layers fit together

1. The estimate is computed first. All money is deterministic arithmetic in the cost engine. Nothing in this document sets a price.
2. DFM and DFA rules read the finished estimate — the 8-bucket split, the operation list, utilisation, OEE, manning — and flag what is out of benchmark for that commodity.
3. Geometry advisors read the measured CAD when there is any: wall thickness, draft angle, section ratio, bend radius. These are process physics, not cost ratios, so they fire whether or not the cost looks odd.
4. Idea levers propose actions, ranked by saving. They are gated by commodity and by what the tool actually knows: a lever built on an assumed volume is downgraded to a confirm-first note rather than an instruction.
5. The headline saving is a root-sum-square of the top three, capped at 40% — never a sum, because levers overlap and adding them overstates what is really available.

1 — Design-for-Manufacture rules (41)

14 rules run on every commodity; 27 are gated to specific processes. The trigger condition is the literal expression from the source — these are the thresholds, not a paraphrase.

1A Universal rules — every commodity

#	Rule / trigger condition	Applies to	Sev.	Saving
1	**Low material utilisation (<60%)** `Fires when: matUtil < 0.60` Material utilisation at ‹matUtil100toFixed0›% is critically low. Over 40% of purchased material becomes waste. **Action:** Improve nesting/billet size; consider near-net-shape process *Why:* All-commodity rules — material utilisation	All commodities	**CRITICAL**	12%
2	**Below-benchmark material utilisation** `Fires when: matUtil >= 0.60 && matUtil < 0.72` Material utilisation at ‹matUtil100toFixed0›% is below the typical 72% benchmark for this commodity. **Action:** Review blank/billet sizing; optimise nesting layout; explore near-net-shape pre-forms *Why:* All-commodity rules — material utilisation	All commodities	**MAJOR**	6%
3	**Low OEE (<70%)** `Fires when: avgOEE < 0.70` Average OEE at ‹avgOEE100toFixed0›% is critically below the 70% threshold. Significant capacity and cost loss. **Action:** Investigate root causes: downtime, changeover, micro-stops. Implement TPM. *Why:* All-commodity rules — OEE	All commodities	**CRITICAL**	15%
4	**Below-target OEE (70–80%)** `Fires when: avgOEE >= 0.70 && avgOEE < 0.80` Average OEE at ‹avgOEE100toFixed0›% is below the 80% target. Improvement opportunity exists. **Action:** Apply OEE improvement toolkit: reduce minor stoppages, optimise changeover, improve quality rate *Why:* All-commodity rules — OEE	All commodities	**MAJOR**	7%
5	**High tooling amortisation (>20% of part cost)** `Fires when: toolPct > 20` Tooling at ‹toolPcttoFixed1›% of part cost is significantly above the 12% benchmark. Volume-sensitive lever. **Action:** Review tool life, increase annual volume, consider modular tooling *Why:* All-commodity rules — tooling	All commodities	**MAJOR**	8%
6	**Elevated tooling amortisation (12–20%)** `Fires when: toolPct > 12 && toolPct <= 20` Tooling at ‹toolPcttoFixed1›% of part cost is above the typical range. Consider volume increase or tooling cost reduction. **Action:** Evaluate family tooling or increase volume to reduce per-part tooling cost *Why:* All-commodity rules — tooling	All commodities	**MINOR**	4%
7	**Overhead burden >18%** `Fires when: oheadPct > 18` Overhead at ‹oheadPcttoFixed1›% of part cost exceeds the 18% benchmark. Factory burden may be inflated. **Action:** Request itemised overhead breakdown; compare tier-1 vs tier-2 structures; explore make-vs-buy *Why:* All-commodity rules — overhead	All commodities	**MAJOR**	6%
8	**Supplier margin >18%** `Fires when: mgnPct > 18` Supplier margin at ‹mgnPcttoFixed1›% exceeds the competitive 18% ceiling. Open negotiation lever available. **Action:** Dual-source or RFQ rebalance *Why:* All-commodity rules — margin	All commodities	**MAJOR**	5%
9	**Packaging cost heavy (>5% of part cost)** `Fires when: pkgPct > 5` Packaging at ‹pkgPcttoFixed1›% of part cost is above the 5% benchmark for engineered parts. **Action:** Evaluate returnable packaging loop; redesign dunnage for pack density; challenge expendable-pack pricing *Why:* All-commodity rules — packaging	All commodities	**MAJOR**	4%

#	Rule / trigger condition	Applies to	Sev.	Saving
10	**Logistics cost heavy (>8% of part cost)** `Fires when: logPct > 8` Logistics at ‹logPcttoFixed1›% of part cost is above the 8% benchmark. **Action:** Review freight mode (sea vs air), consolidation, incoterm and pack density; run the landed-cost comparison *Why:* All-commodity rules — logistics	All commodities	**MAJOR**	5%
11	**One operation dominates — "‹name›"** `Fires when: topOp && topOpShare > 0.6 && opCount >= 3` "‹name›" carries ‹topOpShare100toFixed0›% of the conversion cost across ‹opCount› operations. The part's cost rides on a single process step. **Action:** Focus cycle-time engineering on the governing operation before touching anything else; ask the supplier for its OEE and changeover data *Why:* All-commodity rules — one operation dominates the conversion cost	All commodities	**MAJOR**	8%
12	**Low labour efficiency (<80%)** `Fires when: avgLabEff < 0.80` Average labour efficiency at ‹avgLabEff100toFixed0›% means a fifth of paid labour minutes add no value. **Action:** Line-balance the manual content; remove walking/waiting via cell layout; standard work instructions *Why:* All-commodity rules — labour efficiency	All commodities	**MINOR**	4%
13	**Consumables dominate the material line** `Fires when: consumShare > 0.25` Per-part consumables (cores, patterns, shell, filters) are ‹consumShare100toFixed0›% of the material cost line. **Action:** Rationalise core count and pattern life; evaluate reclaim (shell sand, wax); challenge consumable pricing open-book *Why:* All-commodity rules — consumables share of the material line	All commodities	**MAJOR**	6%
14	**Tooling amortised over less than the annual volume** `Fires when: annualVol > 0 && amortVol > 0 && amortVol < annualVol * 0.9` The tooling is amortised over ‹amortVoltoLocaleString› parts against a stated annual volume of ‹annualVoltoLocaleString› — the piece price carries more tooling per part than a full-year basis would. **Action:** Confirm the amortisation basis with the supplier; align it to programme volume or move tooling to a separate NRE line *Why:* All-commodity rules — amortisation basis vs stated annual volume. A data/commercial check, not a saving claim.	All commodities	**MINOR**	0%

1B Commodity-gated rules

#	Rule / trigger condition	Applies to	Sev.	Saving
15	**Split routing across ‹stations› stations** `Fires when: st.opCount > 4 && st.stations > 2 && !st.consolidated` ‹opCount› machining operations across ‹stations› separate stations. Each station adds fixture cost, handling time, and process variation. **Action:** Quote a multi-axis consolidation against the split routing *Why:* cast_and_machine used to be absent here while the DFA block below was ungated — the same part scored DFM 10/10 and DFA MAJOR for the same fact.	machining, cast and machine	**MAJOR**	10%
16	**Process cost dominates (>40%)** `Fires when: procPct > 40` Process cost at ‹procPcttoFixed1›% of total. Machining conversion cost is unusually high. **Action:** Consider near-net-shape pre-form (casting/forging)	machining, cast and machine	**MAJOR**	8%
17	**Die cost very high (>18%)** `Fires when: toolPct > 18` Die tooling at ‹toolPcttoFixed1›% of part cost is critically high. Volume is insufficient to amortise die investment. **Action:** Increase annual volume, consider family tooling or gravity die as interim solution	casting, cast and machine	**CRITICAL**	12%
18	**Unusually low material content — verify alloy grade** `Fires when: matPct < 35` Material at ‹matPcttoFixed1›% of part cost is below the 35% floor for castings. Verify alloy grade and net weight inputs. **Action:** Validate material weight and alloy price inputs against purchase orders	casting, cast and machine	**MINOR**	0%

#	Rule / trigger condition	Applies to	Sev.	Saving
19	**Low forging material utilisation (<75%)** `Fires when: matUtil < 0.75` Material utilisation at ‹matUtil100toFixed0›% is below the 75% forging benchmark. Flash and scale loss is excessive. **Action:** Optimise preform design to reduce flash; consider closed-die forging with controlled flash land	forging	**MAJOR**	8%
20	**Die cost high (>15%)** `Fires when: toolPct > 15` Forging die cost at ‹toolPcttoFixed1›% of part cost. High tool investment relative to volume. **Action:** Increase annual volume commitment; explore hot-trim tooling consolidation	forging	**MAJOR**	7%
21	**Blank nesting <65%** `Fires when: matUtil < 0.65` Material utilisation at ‹matUtil100toFixed0›% implies poor blank nesting. Over 35% of sheet becomes scrap offcuts. **Action:** Optimise nesting with CAD nesting software; consider part redesign to reduce scrap skeleton	sheet metal, sheet metal fab	**CRITICAL**	15%
22	**Too many forming operations** `Fires when: opCount > 6` ‹opCount› operations detected. Excessive forming stages increase cycle time and die investment. **Action:** Combine operations using progressive or transfer die; review form sequence for efficiency	sheet metal, sheet metal fab	**MAJOR**	8%
23	**Seam welding — distortion risk** `Fires when: hasSeamWeld` Continuous MIG/TIG welding on thin sheet induces heat distortion, residual stress and post-weld straightening cost. **Action:** Prefer spot welds, clinching or self-pierce rivets; if welding is required use stitch welds, a balanced sequence and fixturing *Why:* Continuous welding on sheet → heat distortion + residual stress	sheet metal, sheet metal fab	**MAJOR**	6%
24	**Tooling is ‹toolPcttoFixed0›% of piece cost** `Fires when: toolPct > 40` Hard-tooling NRE dominates the piece cost — the programme volume is too low to amortise a stamping die / press-brake tooling. **Action:** Increase the amortisation volume, or switch to a low-NRE route (laser + press brake) until volume grows *Why:* Tooling-dominated cost → volume too low to amortise hard tooling	sheet metal, sheet metal fab	**MAJOR**	10%
25	**Mould cost >30%** `Fires when: toolPct > 30` Mould/tool cost at ‹toolPcttoFixed1›% of part cost is critically high. Volume is insufficient to justify tooling investment. **Action:** Increase annual volume; consider family/multi-cavity tooling; evaluate aluminium soft tooling for low volumes	injection moulding, blow moulding, thermoforming	**CRITICAL**	12%
26	**Runner/sprue waste >25%** `Fires when: matUtil < 0.75` Material utilisation at ‹matUtil100toFixed0›%. Significant runner and sprue waste. Hot runner system may eliminate waste entirely. **Action:** Evaluate hot runner system to eliminate cold runner waste; increase cavity count	injection moulding, blow moulding, thermoforming	**MAJOR**	6%
27	**Fab complexity driving high process cost** `Fires when: procPct > 60` PCB fabrication process cost at ‹procPcttoFixed1›% of total. High layer count, fine pitch, or tight tolerances are driving cost. **Action:** Review layer stack-up; relax minimum track/space where functional spec allows; consider standard via sizes	pcb fab	**MAJOR**	10%
28	**NRE cost per board high — increase volume** `Fires when: toolPct > 10` NRE/setup cost at ‹toolPcttoFixed1›% per board. Volume is insufficient to amortise non-recurring costs. **Action:** Increase annual volume; evaluate panelisation to maximise boards per setup run	pcb fab	**MAJOR**	8%
29	**High labour content — automate SMT** `Fires when: labPct > 35` Labour at ‹labPcttoFixed1›% of PCBA cost. High manual content indicates through-hole or manual rework operations. **Action:** Convert through-hole components to SMT equivalents; automate selective soldering; reduce manual inspection	pcba	**MAJOR**	12%

#	Rule / trigger condition	Applies to	Sev.	Saving
30	**Many assembly operations** `Fires when: opCount > 6` ‹opCount› assembly operations detected. Complex PCBA with many process steps increases cycle time and defect risk. **Action:** Consolidate assembly stages; review component placement for assembly efficiency	pcba	MAJOR	7%
31	**Cure process cost dominant** `Fires when: procPct > 45` Process cost at ‹procPcttoFixed1›% of total. Moulding and cure cycle time is the primary cost driver. **Action:** Optimise cure recipe (temperature/time); evaluate injection moulding vs compression; increase cavity count	rubber	MAJOR	8%
32	**Labour-intensive layup** `Fires when: labPct > 40` Labour at ‹labPcttoFixed1›% of composite part cost. Manual layup is the dominant cost driver. **Action:** Evaluate ATL/AFP automated fibre placement; consider resin infusion over manual prepreg layup	composites	MAJOR	15%
33	**Long cure cycle cost** `Fires when: procPct > 35` Process cost at ‹procPcttoFixed1›%. Autoclave or oven cure cycle time driving high process cost. **Action:** Evaluate out-of-autoclave (OOA) processes; consider press moulding for high volume; optimise cure schedule	composites	MAJOR	10%
34	**Labour >50% — automation critical** `Fires when: labPct > 50` Labour at ‹labPcttoFixed1›% of harness cost. Manual termination and assembly is the dominant cost element. **Action:** Automate wire cutting, stripping, and terminal crimping; evaluate connector standardisation for robotic assembly	wiring harness	CRITICAL	20%
35	**High complexity harness** `Fires when: opCount > 8` ‹opCount› harness assembly operations. Complex routing and branching increases assembly time and defect risk. **Action:** Standardise connector families; reduce branch points; evaluate sub-harness modularisation	wiring harness	MAJOR	8%
36	**Extrusion conversion cost high (>40%)** `Fires when: procPct > 40` Process cost at ‹procPcttoFixed1›% of total. Die design, puller speed or billet temperature is limiting throughput. **Action:** Review die bearing lengths and port design; evaluate a multi-hole die; optimise billet temperature and ram speed for the alloy	extrusion	MAJOR	8%
37	**High start-up and offcut scrap (<80% utilisation)** `Fires when: matUtil < 0.80` Material utilisation at ‹matUtil100toFixed0›%. Butt ends, start-up lengths and cut-to-length offcuts exceed the 80% extrusion benchmark. **Action:** Optimise cut plan against the order lengths; recycle offcuts closed-loop; negotiate a scrap credit at the billet price	extrusion	MAJOR	5%
38	**Oven cycle dominates (>50% of part cost)** `Fires when: procPct > 50` Process cost at ‹procPcttoFixed1›% of total. The oven cook/cool cycle is the governing cost element. **Action:** Increase arm loading density (more moulds per arm); optimise cook control (internal-air temperature control); review wall thickness — cycle scales with wall	rotational moulding	MAJOR	8%
39	**Many secondary operations after moulding** `Fires when: opCount > 4` ‹opCount› operations on a rotomoulded part. Trimming, drilling and fitting content is adding labour beyond the moulding itself. **Action:** Mould-in features (inserts, kiss-offs, apertures with moulded witness lines) to delete downstream operations	rotational moulding	MINOR	4%
40	**Paint material heavy — transfer efficiency lever** `Fires when: matPct > 40` Paint material at ‹matPcttoFixed1›% of cost. At typical 60% transfer efficiency, 4 of every 10 litres bought ends in the booth filters. **Action:** Improve transfer efficiency (electrostatics, robot path optimisation); review film-build spec vs requirement; reclaim/recirculate where the chemistry allows	painting	MAJOR	8%

#	Rule / trigger condition	Applies to	Sev.	Saving
41	**High facility overhead** `Fires when: oheadPct > 20` Overhead at ‹oheadPcttoFixed1›% of part cost. Paint shop or BIW facility burden rate is elevated. **Action:** Review facility utilisation; consolidate product mix to improve line efficiency; negotiate toll-painting rates	painting, biw assembly	**MAJOR**	6%

2 — Design-for-Assembly rules (11)

Assembly and handling checks in the Boothroyd-Dewhurst tradition: does the routing imply more stations than the part needs, is manning higher than the operation warrants, is the part count or fastening scheme adding handling that design could remove.

#	Rule / trigger condition	Applies to	Sev.	Saving
1	**‹stations› separate stations imply multiple setups/transfers** `Fires when: st.opCount > 4 && st.stations > 2 && !st.consolidated` ‹opCount› machining operations across ‹stations› stations require multiple setups or work transfers, increasing total process cycle time. **Action:** Quote multi-axis consolidation, combined tooling, or part consolidation against this routing	All commodities	**MAJOR**	8%
2	**Multi-operation routing already consolidated** `Fires when: st.opCount > 4 && st.consolidated` ‹opCount› operations run across ‹stations› station(s) **Action:** Use the routing comparison in the derivation trace as negotiation evidence	All commodities	**OPPORTUNITY**	0%
3	**High manual content — assess robot/automation feasibility** `Fires when: opCount > 3 && labPct > 30` ‹opCount› operations with ‹labPcttoFixed1›% labour content indicates significant manual intervention in the process. **Action:** Conduct automation feasibility study; evaluate cobot integration for repetitive assembly tasks	All commodities	**MAJOR**	12%
4	**Low OEE indicates manual pacing or frequent stoppages** `Fires when: avgOEE < 0.75` Average OEE at ‹avgOEE100toFixed0›% suggests manual pacing, frequent micro-stops, or high changeover time between parts. **Action:** Implement line balancing study; standardise operator work cycles; introduce poka-yoke to reduce defect-driven stoppages	All commodities	**MAJOR**	8%
5	**Labour-dominated assembly — priority for fixture/automation** `Fires when: (commodity === 'wiring_harness' || commodity === 'pcba') && labPct > 45` Labour at ‹labPcttoFixed1›% in a ‹commodityreplace_g› context is critically high. Manual assembly dominates cost. **Action:** Invest in dedicated assembly fixtures; evaluate automated soldering, crimping, and testing equipment	wiring harness, pcba	**CRITICAL**	18%
6	**Multiple fixturing operations; consider pallet systems** `Fires when: (commodity === 'machining' || commodity === 'forging') && st.stations > 2` ‹opCount› operations across ‹stations› stations in ‹commodityreplace_g› imply multiple fixturing steps. Pallet systems can reduce handling time. **Action:** Evaluate tombstone/pallet machining systems for batch processing without re-fixturing	machining, forging	**MINOR**	5%
7	**Assess fastener standardisation** `Fires when: opCount > 4 && labPct > 20` Multiple operations with labour content suggest fastener variety may be adding assembly complexity and tool change time. **Action:** Standardise fastener type and drive; reduce fastener count through snap-fit or weld; evaluate fastener family rationalisation	All commodities	**OPPORTUNITY**	4%
8	**Inspection is a separate manual step** `Fires when: opNamesAll.some(n => RX_INSPECT.test(n)) && inspConvShare > 0.05` Standalone inspection/test operations carry ‹inspConvShare100toFixed1›% of part cost. Capable processes should be verifying in-line, not at a separate station. **Action:** Move to in-line gauging / on-machine probing; SPC-based skip-lot once capability is demonstrated *Why:* Boothroyd-flavoured checks read from the operation list itself.	All commodities	**MINOR**	3%
9	**Manual finishing after the primary process** `Fires when: opNamesAll.some(n => RX_REWORK.test(n))` Deburring/fettling/rework appears as its own operation — a symptom of an upstream condition (tool wear, gating, die state) being paid for by hand, every part. **Action:** Fix the burr at source (tooling discipline, gate redesign); where finishing must remain, tumble/vibratory beats manual	All commodities	**MINOR**	3%

#	Rule / trigger condition	Applies to	Sev.	Saving
10	**High manning (‹maxManningtoFixed1› operators on one operation)** `Fires when: maxManning >= 2` At least one operation runs with ‹maxManningtoFixed1› operators. Multi-manned stations are the strongest automation candidates on any part. **Action:** Line-balance or automate the multi-manned station first; check whether the second operator exists only to load/unload	All commodities	**MAJOR**	6%
11	**Labour time exceeds the machine cycle** `Fires when: labPct > 15 && (input.operations ?? []).some(o => (o.labourTimeHr ?? 0) > (o.cycleTimeHr ?? 0) * 1.2 && (o.cycleTimeHr ?? 0) > 0)` On at least one operation the charged labour minutes exceed the machine cycle — manual work content is running beyond the machine-paced window. **Action:** Move the excess manual content offline (pre-kitting, subassembly) or in parallel with the cycle; review what the operator actually does during the cycle	All commodities	**MINOR**	4%

3 — Geometry advisors (10 modules, 75 checks)

One module per process family. Each reads measured geometry and the process reference table for the chosen route, then reports what the physics will not allow — a wall below the minimum the process can fill, a section ratio that will draw shrinkage porosity, a bend radius below the material minimum. These are independent of cost: a part can be cheap and still be unmakeable.

Blow — `analyseBlowDFM()` · 6 checks

#	Rule / trigger condition	Applies to	Sev.
1	**Blow-up ratio ‹inputsblowUpRatiotoFixed1› exceeds ~3.5** `Fires when: inputs.blowUpRatio !== undefined && inputs.blowUpRatio > 3.5` A high BUR over-stretches the parison — walls thin unevenly, corners run weak and blow-out risk rises. **Action:** Keep BUR ≤ 3:1 (≤3.5 max); use a larger parison/die head or parison programming to redistribute wall. *Why:* 1. Blow-up ratio too high → thin, weak walls / blow-out.	blow	process
2	**Wall ‹t› mm is very thin for blow moulding** `Fires when: t > 0 && t < 0.5` Sub-0.5 mm blown walls tear, pinhole and fail top-load; wall control across the part is poor. **Action:** Increase nominal wall to ≥ 0.6–0.8 mm, or add parison wall-thickness programming (axial) to thicken load-bearing zones. *Why:* 2. Thin wall.	blow	MAJOR
3	**Corner radius ‹minCornerRadiusMm› mm below 2×wall (‹2ttoFixed1› mm)** `Fires when: inputs.minCornerRadiusMm !== undefined && t > 0 && inputs.minCornerRadiusMm < 2 * t` Sharp corners are where the parison stretches most — they thin to a fraction of nominal wall and crack. **Action:** Radius external corners to ≥ 2–3×wall; sharp detail should be on the least-stretched face. *Why:* 3. Corner thinning.	blow	MAJOR
4	**Parison L/D ‹inputsparisonLtoDtoFixed0› — sag / weight drift** `Fires when: process === 'ebm' && inputs.parisonLtoD !== undefined && inputs.parisonLtoD > 8` Long parisons sag and thin under their own weight before clamp, drifting wall distribution shot-to-shot. **Action:** Use an accumulator head, faster extrusion, or parison programming; consider splitting into shorter parts. *Why:* 4. Parison sag on long EBM parts.	blow	MINOR
5	**Handle / weld line in a load path** `Fires when: inputs.handleOrWeldLine` The pinch weld behind a handle or offset neck is the weakest line on an EBM part and a common failure origin. **Action:** Thicken and radius the pinch-off, keep the weld out of the top-load path, and validate drop/burst on the weld. *Why:* 5. Weld/pinch line on a handled part.	blow	MINOR
6	**Tolerance ±‹toleranceMm› mm tight for ‹processtoUpperCase› (~±‹cap› mm)** `Fires when: inputs.toleranceMm < cap` Blown walls are not dimensionally tight; only tool-formed features (neck/finish) hold close tolerance. *Why:* 6. Tight tolerance — process capability by type.	blow	MINOR

Casting — `analyseCastingDFM()` · 7 checks

Reference tables: `CASTING_PROCESS_REFERENCE`

#	Rule / trigger condition	Applies to	Sev.
1	**Wall ‹minWallThicknessMm› mm below ‹process› minimum ‹minWallMm› mm** `Fires when: inputs.minWallThicknessMm < ref.minWallMm` Thin sections freeze before the cavity fills — cold shuts, misruns and non-fill scrap. **Action:** Thicken to ≥ ‹minWallMm› mm, add flow ribs, raise metal/die temperature, or move to a process with lower min wall (investment ‹minWallMm› mm). *Why:* Wall thickness below process minimum → non-fill / cold shut / misrun.	casting	CRITICAL

#	Rule / trigger condition	Applies to	Sev.
2	**Non-uniform sections (heavy:thin ≈ ‹ratiotoFixed1›:1)** `Fires when: ratio > 3` Heavy sections solidify last and shrink, drawing porosity/sink at the hot spot. **Action:** Even out wall thickness, core out heavy bosses, add fillets to taper transitions, or feed the hot spot with a riser/chill. *Why:* Section ratio → shrinkage porosity at hot spots.	casting	**MAJOR**
3	**Draft ‹draftAngleDeg›° below recommended ‹draftDegMin›°** `Fires when: inputs.draftAngleDeg < ref.draftDegMin` Insufficient draft galls the die/pattern, raises ejection force and accelerates tool wear (or breaks the sand mould). **Action:** Add ≥ ‹draftDegMin›° draft on all as-cast vertical walls; deep pockets need more. Investment casting can run near-zero draft if this is fixed. *Why:* Draft below process minimum → ejection / die wear / pattern draw.	casting	tooling
4	**Isolated heavy section / boss** `Fires when: inputs.hasIsolatedHeavySections` A thick mass with no feed path shrinks internally — subsurface porosity that only shows after machining. **Action:** Connect the heavy section to a feeder, add a chill to reverse the freeze direction, or hollow/core the mass. *Why:* Isolated heavy sections → shrinkage porosity needing feed.	casting	**MAJOR**
5	**Leak-tight HPDC without gas-porosity mitigation** `Fires when: inputs.pressureTight && (inputs.process === 'hpdc' || inputs.process === 'megacasting') && !inputs.porosityMitigated` Conventional HPDC entraps air — gas porosity fails leak tests and cannot be welded/heat-treated to full spec. **Action:** Specify vacuum-assist HPDC, or add a resin impregnation step; both carry a cost adder (see secondary-process estimator). *Why:* Pressure-tight parts without porosity mitigation.	casting	**MAJOR**
6	**Sharp internal corners** `Fires when: inputs.hasSharpInternalCorners` Sharp re-entrant corners concentrate stress and are hot-tear initiation sites during solidification. **Action:** Add generous fillets (radius ≥ adjoining wall thickness) at all internal corners. *Why:* Sharp internal corners → hot tears / stress raisers.	casting	**MINOR**
7	**Machining stock ‹machiningStockMm› mm exceeds near-net target** `Fires when: inputs.machiningStockMm !== undefined && inputs.machiningStockMm > 2` Every mm of all-over stock adds pour weight, machining time and swarf that is only worth scrap value. **Action:** Tighten as-cast tolerance and cut stock toward 0.5–1.0 mm on functional faces only; leave non-functional faces as-cast. *Why:* Excess machining stock → material + cycle waste (near-net opportunity).	casting	**OPPORTUNITY**

Extrusion — `analyseExtrusionDFM()` · 8 checks

#	Rule / trigger condition	Applies to	Sev.
1	**Min wall ‹minWallMm› mm below ~‹floor› mm for ‹process›** `Fires when: inp.minWallMm !== undefined && inp.minWallMm > 0 && inp.minWallMm < floor` Very thin sections draw-down unpredictably and tear; die lands cannot be finished thin enough. **Action:** Raise the thinnest section to ≥ ‹floor› mm, or split into two co-extruded layers. *Why:* 1. Minimum wall — dies cannot hold very thin extruded sections cleanly.	extrusion	**MAJOR**
2	**Wall ratio ‹inpmaxWallMminpminWallMmtoFixed1›:1 (>3:1) — differential cooling** `Fires when: inp.minWallMm && inp.maxWallMm && inp.minWallMm > 0 && inp.maxWallMm / inp.minWallMm > 3` Thick and thin sections in the same profile cool at different rates, causing bow, twist and sink marks. **Action:** Even out wall thickness (target <3:1); core out thick sections or add internal ribs instead of solid mass. *Why:* 2. Wall-thickness uniformity — uneven walls cool at different rates → warp/bow.	extrusion	**MAJOR**

<table>
<tr><th>#</th><th>Rule / trigger condition</th><th>Applies to</th><th>Sev.</th></tr>
<tr><td>3</td><td>Internal radius ‹minInternalRadiusMm› mm sharp vs wall
<code>Fires when: inp.minInternalRadiusMm !== undefined && inp.wallThicknessMm && inp.minInternalRadiusMm < 0.5 * inp.wallThicknessMm</code>
Sharp internal corners cause flow hesitation and become fatigue/stress concentrators in the profile.
Action: Radius internal corners to ≥ 0.5×wall; generous radii improve flow balance and part strength at no tooling penalty.
Why: 3. Sharp internal corners — flow hesitation + stress raiser.</td><td>extrusion</td><td>MINOR</td></tr>
<tr><td>4</td><td>Tolerance ±‹toleranceMm› mm tighter than extrudable ±‹achievabletoFixed2› mm
<code>Fires when: inp.toleranceMm < achievable</code>
Extrusion drifts with melt temperature, line speed and die swell; sub-process tolerances drive scrap and 100% gauging.
Action: Relax the tolerance to the extrusion band, add downstream calibration/sizing, or post-machine the critical feature.
Why: 4. Tolerance realism — extrusion is a loose-tolerance process.</td><td>extrusion</td><td>MAJOR</td></tr>
<tr><td>5</td><td>‹hollowChambers› hollow chambers — sizing complexity
<code>Fires when: inp.hollowChambers !== undefined && inp.hollowChambers >= 2</code>
Multi-chamber hollow profiles need internal cooling/vacuum and precise mandrel support; each chamber adds die and calibration cost.
Action: Minimise chamber count; ensure each hollow has vacuum access and balanced wall for even cooling.
Why: 5. Hollow / multi-lumen chambers — cooling & sizing complexity.</td><td>extrusion</td><td>MINOR</td></tr>
<tr><td>6</td><td>‹layers›-layer co-extrusion
<code>Fires when: inp.layers !== undefined && inp.layers >= 4</code>
Each layer adds an extruder, a feed manifold and interfacial-instability risk; >5 layers rarely justified outside barrier packaging.
Action: Confirm each layer earns its function (barrier, tie, cap); collapse tie/cap layers where properties allow.
Why: 6. Co-extrusion layer count.</td><td>extrusion</td><td>MINOR</td></tr>
<tr><td>7</td><td>Unsupported projection ‹unsupportedProjectionMm› mm (>15×wall)
<code>Fires when: inp.unsupportedProjectionMm !== undefined && inp.wallThicknessMm && inp.unsupportedProjectionMm > 15 * inp.wallThicknessMm</code>
Long thin legs/fins sag under their own weight before the calibrator captures them, going out of position.
Action: Thicken the projection, shorten it, or add a tie-in web so the calibrator can hold its position.
Why: 7. Long unsupported thin projection — droops off the die.</td><td>extrusion</td><td>MINOR</td></tr>
<tr><td>8</td><td>‹famtoUpperCase› die swell ~‹estimateDieSwellPctfam›% — allow die-tuning trials
<code>Fires when: (fam === 'pe' || fam === 'pp' || fam === 'tpe') && (inp.process === 'profile' || inp.process === 'profile-complex')</code>
High-swell polymers need the die profile shrunk vs the target section and several tuning trials at first-off.
Action: Budget die-swell tuning scrap; consider a lower-swell grade or a filled compound for dimensionally critical profiles.
Why: 8. Die-swell advisory for high-swell families on tight profiles.</td><td>extrusion</td><td>OPPORTUNITY</td></tr>
</table>

Forging — `analyseForgingDFM()` · 7 checks

Reference tables: `FORGING_PROCESS_REFERENCE`, `FORGING_FLOW_STRESS_MPA`, `FORGING_DIE_LIFE_BASE`, `FORGING_HEAT_KWH_PER_KG`

<table>
<tr><th>#</th><th>Rule / trigger condition</th><th>Applies to</th><th>Sev.</th></tr>
<tr><td>1</td><td>Web ‹minWebThicknessMm› mm below ‹process› minimum ‹minWebMm› mm
<code>Fires when: inputs.minWebThicknessMm < ref.minWebMm</code>
Thin webs cool fast and resist metal flow — incomplete fill, forging laps and excessive die pressure/wear.
Action: Thicken the web to ≥ ‹minWebMm› mm, or move to a lower-web process (precision ‹minWebMm› mm / cold ‹minWebMm› mm).
Why: Web/rib below process minimum → non-fill / laps / die overload.</td><td>forging</td><td>CRITICAL</td></tr>
</table>

#	Rule / trigger condition	Applies to	Sev.
2	**Draft ‹draftAngleDeg›° below recommended ‹draftDegMin›°** `Fires when: inputs.draftAngleDeg < ref.draftDegMin` Insufficient die draft prevents clean part release — galling, ejection cracks and rapid die wear. **Action:** Add ≥ ‹draftDegMin›° draft on impression walls (deep cavities need 5–7°). Precision/warm forging can run reduced draft with ejectors. *Why:* Draft below process minimum → part sticks in die / ejection damage.	forging	tooling
3	**Sharp fillet radius ‹filletRadiusMm› mm** `Fires when: inputs.filletRadiusMm !== undefined && inputs.filletRadiusMm < 3 && inputs.process !== 'cold-forming'` Small corner/fillet radii choke metal flow into the impression, causing laps/cold shuts and concentrating die stress. **Action:** Open fillet radii to ≥ 3–5 mm; generous radii improve die fill and die life and reduce forging load. *Why:* Sharp fillets → poor metal flow, laps, stress raisers.	forging	**MAJOR**
4	**Deep thin rib (height:thickness ≈ ‹inputsribHeightToThicknesstoFixed1›:1)** `Fires when: inputs.ribHeightToThickness !== undefined && inputs.ribHeightToThickness > 4` Tall thin ribs are the hardest features to fill and the first to lap or under-fill. **Action:** Reduce rib aspect ratio (<4:1), add draft and radius, or split into a two-blow preform + finish sequence. *Why:* Deep thin ribs → non-fill.	forging	**MAJOR**
5	**Grain flow not aligned with primary load path** `Fires when: inputs.grainFlowAligned === false` The core advantage of forging is continuous grain flow; cutting across it in a highly-loaded region forfeits fatigue life vs a bar-stock machining. **Action:** Reorient the parting line/preform so forged fibre follows the main stress direction; avoid machining through end-grain on loaded faces. *Why:* Grain flow not aligned with load path → fatigue underperformance.	forging	**MAJOR**
6	**Parting line on a loaded/sealing face** `Fires when: inputs.partingLineOnCriticalFace` Flash line and die mismatch land on a functional surface — extra machining and a fatigue/sealing risk. **Action:** Move the parting line to a non-critical face, or add clean-up machining stock only where the flash line falls. *Why:* Parting line on a critical face → flash/mismatch on a functional surface.	forging	**MINOR**
7	**Machining stock ‹machiningStockMm› mm exceeds near-net target** `Fires when: inputs.machiningStockMm !== undefined && inputs.machiningStockMm > 3 && inputs.process !== 'open-die'` Excess all-over envelope adds input weight, forging load and machining — swarf worth only scrap value. **Action:** Tighten the forging to near-net (precision route) and cut stock toward 1–2 mm on functional faces only. *Why:* Excess machining stock → near-net opportunity.	forging	**OPPORTUNITY**

Injection — `analyseInjectionDFM()` · 10 checks

#	Rule / trigger condition	Applies to	Sev.
1	**Wall ‹t› mm exceeds ‹max› mm — thick-section sink & cycle penalty** `Fires when: t > 0 && t > win.max` Thick walls cool as the square of thickness, stretching cycle time, and shrink internally to form sinks and voids. **Action:** Core out the section to a uniform 2–3 mm wall with ribs for stiffness; add gas-assist only if a heavy section is unavoidable. *Why:* 1. Nominal wall thickness inside the mouldable window.	injection	**MAJOR**
2	**Wall ‹t› mm below ‹min› mm — short-shot / high-pressure risk** `Fires when: t > 0 && t < win.min` Walls thinner than the resin can reliably fill cause short shots, high injection pressure and premature freeze-off. **Action:** Increase wall to ≥ ‹min› mm, add flow leaders, or move the gate closer to the thin region. *Why:* 1. Nominal wall thickness inside the mouldable window.	injection	**MAJOR**
3	**Wall varies ‹ratiotoFixed1›× (‹minWallMm›–‹maxWallMm› mm) — differential shrink** `Fires when: ratio > 2.0` **Action:** Hold wall within ±25% of nominal; blend transitions over ≥3×wall and relocate mass into ribs rather than solid sections. *Why:* 2. Wall uniformity — sink & warp from thick/thin transitions.	injection	geometry

#	Rule / trigger condition	Applies to	Sev.
4	**Rib ‹inputsribThicknessRatio100toFixed0›% of wall — visible sink** `Fires when: inputs.ribThicknessRatio !== undefined && inputs.ribThicknessRatio > 0.6` A rib thicker than ~60% of the nominal wall pulls a sink mark on the opposite (show) face as it shrinks. **Action:** Thin ribs to 40–60% of nominal wall; if strength needs more, use multiple thin ribs or a gusset rather than one thick rib. *Why:* 3. Rib thickness — sink on the show face.	injection	geometry
5	**Boss wall ‹inputsbossWallRatio100toFixed0›% of nominal — sink risk** `Fires when: inputs.bossWallRatio !== undefined && inputs.bossWallRatio > 0.6` A boss joined to the wall at >60% thickness sinks and traps gas at the base. **Action:** Set boss outer wall to ~60% of nominal, connect with ribs, and cored to a uniform thickness; add a base radius. *Why:* 4. Boss wall — sink & voids.	injection	**MINOR**
6	**Draft ‹draftAngleDeg›° below ‹minDraft›° minimum‹value›** `Fires when: inputs.draftAngleDeg < minDraft` **Action:** Add ≥ ‹minDraft›° draft to all vertical faces (more for deep or textured walls); increase toward 5° for heavy grain. *Why:* 5. Draft angle — ejection drag / scuffing.	injection	geometry
7	**‹undercutCount› undercut‹undercutCount1s› need side-actions / lifters** `Fires when: inputs.undercutCount !== undefined && inputs.undercutCount > 0` Each undercut adds a slide or lifter — raising tool cost, maintenance and cycle time, and capping cavitation. **Action:** Design out undercuts with pass-through cores / bump-offs where the resin allows, or consolidate them onto one moving side. *Why:* 6. Undercuts — side-action / lifter tooling cost.	injection	tooling
8	**Flow-length/wall ‹ratiotoFixed0› exceeds ~‹maxRatio› (L/t)** `Fires when: ratio > maxRatio` Beyond the resin\'s L/t limit the melt freezes before filling — short shots, high pressure and weak weld lines. **Action:** Add gates to shorten each flow path, thicken flow leaders, or select a higher-flow (higher-MFI) grade. *Why:* 7. Flow length vs wall — fill / short-shot.	injection	**MAJOR**
9	**Weld/knit line on a cosmetic or load-bearing face** `Fires when: inputs.weldLineOnCriticalFace` Weld lines are visible witness marks and carry 10–60% less strength — a liability on show surfaces and structural regions. **Action:** Relocate the gate to move the weld line off the critical area, raise melt/mould temperature, or add an overflow well. *Why:* 8. Weld line on a critical face.	injection	**MAJOR**
10	**Tolerance ±‹toleranceMm› mm is tight for moulding** `Fires when: inputs.toleranceMm !== undefined && inputs.toleranceMm > 0 && inputs.toleranceMm < 0.10` **Action:** Relax to ±0.1–0.2 mm where function allows, datum critical features together, or add a post-mould machining step only on the critical dimension. *Why:* 9. Tight tolerance for moulding.	injection	tolerance

Lamination — `analyseLaminationDFM()` · 6 checks

Reference tables: `LAMINATION_ANNEAL_KWH_PER_KG`, `LAMINATION_COATING_GBP_PER_KG`

#	Rule / trigger condition	Applies to	Sev.
1	**Tooth/slot width ‹minToothWidthMm› mm below material thickness ‹t› mm** `Fires when: inputs.minToothWidthMm !== undefined && t > 0 && inputs.minToothWidthMm < t` Punching a feature narrower than the strip thickness breaks slender punches and burrs the lamination edge. **Action:** Keep tooth/slot widths ≥ 1–1.5×thickness, or notch (rather than progressive-blank) the finest features. *Why:* 1. Tooth / slot width vs thickness — punch breakage on narrow features.	lamination	**MAJOR**
2	**Bridge/back-iron ‹minBridgeWidthMm› mm below 1.5×thickness** `Fires when: inputs.minBridgeWidthMm !== undefined && t > 0 && inputs.minBridgeWidthMm < 1.5 * t` Very thin bridges distort during blanking and stacking and saturate magnetically. **Action:** Widen bridges to ≥ 1.5–2×thickness; check magnetic saturation of the narrow flux path. *Why:* 2. Back-iron bridge width.	lamination	**MINOR**
3	**Air-gap tolerance ±‹airGapToleranceMm› mm is very tight** `Fires when: inputs.airGapToleranceMm !== undefined && inputs.airGapToleranceMm > 0 && inputs.airGapToleranceMm < 0.02` Sub-20µm bore/air-gap control fights die wear and stack build-up variation; drives scrap and cogging-torque spread. **Action:** Hold ±0.02–0.03 mm from the progressive die; for tighter, add a final bore grind/hone or bond+machine the stack. *Why:* 3. Air-gap / bore tolerance.	lamination	**MAJOR**
4	**No stress-relief anneal planned** `Fires when: inputs.stressReliefAnneal === false` Blanking work-hardens the cut edge and raises core loss 5–20%; skipping the anneal leaves efficiency on the table. **Action:** Add a stress-relief anneal (~800°C, N■/H■) for efficiency-critical motors; semi-processed grades require it. *Why:* 4. Stress-relief anneal recommendation (loss recovery).	lamination	**OPPORTUNITY**
5	**Laser stack welding shorts edge laminations** `Fires when: inputs.stackMethod === 'laser-weld'` OD welds electrically bridge a few laminations, adding local eddy-current loss — usually acceptable, but not for the lowest-loss designs. **Action:** For high-efficiency EV traction prefer interlock or backlack bonding; keep welds short and off high-flux regions. *Why:* 5. Laser weld magnetic short.	lamination	**MINOR**
6	**Thin-gauge (≤0.27 mm) handling & burr control** `Fires when: inputs.thinGauge` Thin electrical steel burrs and warps easily and wears notch tooling fast, degrading stacking factor and loss. **Action:** Maintain tight punch-die clearance (~5–8% of thickness), regrind tooling frequently, and monitor burr < 8% of thickness. *Why:* 6. Thin-gauge handling / burr.	lamination	**MINOR**

Roto — `analyseRotoDFM()` · 6 checks

#	Rule / trigger condition	Applies to	Sev.
1	**Wall ‹t› mm below roto minimum ~‹minWall› mm** `Fires when: t > 0 && t < minWall` Roto has no pressure to pack thin walls; below ~1.5–2 mm you get pinholes, incomplete sintering and blow-holes. **Action:** Increase nominal wall to ≥ ‹minWall› mm; roto naturally holds a uniform wall so add thickness globally rather than locally. *Why:* 1. Wall too thin to control in roto.	roto	**MAJOR**
2	**Internal radius ‹minInternalRadiusMm› mm below ~3×wall (‹3ttoFixed1› mm)** `Fires when: inputs.minInternalRadiusMm !== undefined && t > 0 && inputs.minInternalRadiusMm < 3 * t` Powder bridges sharp internal corners and starves them — the outside corner runs thin and weak. **Action:** Open internal radii to ≥ 3×wall (ideally 4–5×); generous radii are essentially free in roto tooling. *Why:* 2. Sharp internal corners → powder bridging / thinning.	roto	**MAJOR**
3	**Draft ‹draftAngleDeg›° below 1° minimum** `Fires when: inputs.draftAngleDeg !== undefined && inputs.draftAngleDeg < 1` Parts shrink onto male tool features; too little draft makes demoulding slow and risks tearing. **Action:** Add ≥ 1–2° draft on side walls (more on textured surfaces); consider ejection aids on deep draws. *Why:* 3. Draft for demould.	roto	tooling
4	**Flat unsupported span ‹flatUnsupportedSpanMm› mm — warpage risk** `Fires when: inputs.flatUnsupportedSpanMm !== undefined && inputs.flatUnsupportedSpanMm > 300` Large flat panels warp and sink on cooling because roto walls cool unevenly with no packing pressure. **Action:** Add ribs, domes, kiss-offs or a slight crown (≥ 1.5% of span) to stiffen large flats. *Why:* 4. Large flat unsupported span → warpage.	roto	**MAJOR**
5	**Enclosed hollow with no vent** `Fires when: inputs.enclosedNoVent` Trapped air expands in the oven and contracts on cooling — the part balloons then collapses/dimples without a vent. **Action:** Add a PTFE-lined vent tube sized to the enclosed volume at the highest point; every closed roto cavity needs venting. *Why:* 5. Enclosed volume with no vent.	roto	**CRITICAL**
6	**Double-wall / kiss-off feature** `Fires when: inputs.doubleWallKissOff` Kiss-offs need controlled tool gap (~3–4×wall) to fuse both walls; too large and they do not knit, too small and they thin. **Action:** Target a kiss-off gap of ~3.5×nominal wall and validate weld strength; add a draft so the mating faces release. *Why:* 6. Double-wall / kiss-off feasibility note.	roto	**MINOR**

Rubber — `analyseRubberDFM()` · 8 checks

Reference tables: RUBBER_CURE_BASE_SEC

#	Rule / trigger condition	Applies to	Sev.
1	**Wall ‹t› mm is very thin for moulded rubber** `Fires when: t > 0 && t < 1.0` Sub-1 mm sections are hard to fill before scorch and tear on demould; flash-to-part ratio rises. **Action:** Increase wall to ≥ 1–1.5 mm, or move to LSR injection (fills thin sections) / die-cut sheet. *Why:* 1. Thin wall — incomplete fill / backrind.	rubber	**MAJOR**
2	**Thick section ‹t› mm — long cure & undercure risk** `Fires when: t > 12` Cure time scales with thickness²; thick rubber undercures at the centre (reversion/porosity) and stretches cycle time. **Action:** Core out heavy sections, use a peroxide/efficient cure system, step-cure, or add a post-cure. *Why:* 2. Thick section — long cure + undercure/porosity risk.	rubber	**MAJOR**

#	Rule / trigger condition	Applies to	Sev.
3	**Wall varies ‹ratiotoFixed1›× — uneven cure/shrink** `Fires when: ratio > 3` Thick and thin sections in one part cure at different rates — thin over-cures while thick under-cures. **Action:** Even the wall out (≤2–3× variation); blend transitions; place the gate to balance fill. *Why:* 3. Wall variation — non-uniform cure/shrink.	rubber	**MINOR**
4	**Draft ‹draftAngleDeg›° is minimal** `Fires when: inputs.draftAngleDeg !== undefined && inputs.draftAngleDeg < 0.5` Low draft slows demould and risks tearing on deep or high-durometer parts. **Action:** Add ≥ 0.5–1° draft on deep walls; harder compounds (>70 ShA) need more. *Why:* 4. Draft — demould drag (rubber flexes, so lower severity).	rubber	**MINOR**
5	**Parting/flash line on a sealing face** `Fires when: inputs.flashLineOnSealingFace` Flash and parting-line mismatch on a dynamic/static sealing surface cause leaks and inspection rejects. **Action:** Move the parting line off the seal bead; use a flashless/insert-trim tool or post-mould deflash on that face. *Why:* 5. Flash line on a sealing/functional face.	rubber	**MAJOR**
6	**‹undercutCount› undercut‹undercutCount1s›** `Fires when: inputs.undercutCount !== undefined && inputs.undercutCount > 0` Rubber can bump/strip off modest undercuts, but deep ones need split/collapsible tooling — more cost and cycle. **Action:** Keep undercuts shallow enough to strip (elastic), or accept split-tool cost for deep features. *Why:* 6. Undercuts — bump-off vs tooling.	rubber	tooling
7	**Metal/fabric insert moulding** `Fires when: inputs.metalInsert` Inserts need pre-treatment (grit-blast + adhesive primer), accurate positioning and add load/unload time. **Action:** Design robust insert location, spec the bonding system (Chemlok-type), and add primer + handling cost. *Why:* 7. Insert moulding.	rubber	**OPPORTUNITY**
8	**Tolerance ±‹toleranceMm› mm is tight for rubber** `Fires when: inputs.toleranceMm !== undefined && inputs.toleranceMm > 0 && inputs.toleranceMm < 0.10` Elastomers shrink on cure and creep/compression-set in service — sub-0.1 mm tolerances fight the material, not just the tool. **Action:** Use RMA/ISO 3302 rubber tolerance classes; reserve tight tolerances for bonded metal features, not the rubber itself. *Why:* 8. Tight tolerance vs rubber elasticity.	rubber	tolerance

Sheet Metal — `analyseSheetMetalDFM()` · 7 checks

#	Rule / trigger condition	Applies to	Sev.
1	**Bend radius ‹minBendRadiusMm› mm below minimum ‹minRtoFixed1› mm (‹factor›×t)** `Fires when: inputs.minBendRadiusMm < minR` Bending tighter than the material minimum cracks the outer fibre and raises springback. **Action:** Open the inside radius to ≥ ‹minRtoFixed1› mm, orient the bend across the grain, or specify a more formable temper/grade. *Why:* 1. Bend radius vs thickness — cracking / fracture.	sheet metal	geometry
2	**Hole Ø‹minHoleDiameterMm› mm below minimum ‹minDtoFixed1› mm** `Fires when: inputs.minHoleDiameterMm < minD` Punching a hole smaller than material thickness breaks punches and burrs the edge; laser is slow/pierce-heavy at that size. **Action:** Increase hole diameter to ≥ ‹minDtoFixed1› mm, or drill as a secondary op. *Why:* 2. Hole diameter vs thickness — punch breakage.	sheet metal	**MAJOR**
3	**Feature-to-edge ‹minFeatureToEdgeMm› mm below 2×t (‹2ttoFixed1› mm)** `Fires when: inputs.minFeatureToEdgeMm !== undefined && t > 0 && inputs.minFeatureToEdgeMm < 2 * t` Holes/slots too close to an edge or bend tear out or distort during forming. **Action:** Keep holes ≥ 2×t from edges and ≥ 2.5×t + radius from a bend line. *Why:* 3. Feature-to-edge / hole-to-bend distance — tear-out & distortion.	sheet metal	**MAJOR**

#	Rule / trigger condition	Applies to	Sev.
4	**Tolerance ±‹toleranceMm› mm is tight for sheet metal** `Fires when: inputs.toleranceMm !== undefined && inputs.toleranceMm < 0.10` Sub-0.1 mm tolerances on formed features fight springback and fixturing variation — driving inspection and scrap. **Action:** Relax to ±0.1–0.2 mm where function allows, or add a coining/restrike/machining step only on the critical feature. *Why:* 4. Tight tolerance for sheet.	sheet metal	tolerance
5	**Weld length ‹weldLengthM› m per part — distortion risk** `Fires when: inputs.weldLengthM !== undefined && inputs.weldLengthM > 0.5` Continuous welding on thin sheet induces heat distortion and residual stress. **Action:** Use stitch/skip welds, balanced weld sequence and fixturing; consider spot welding or clinching/SPR instead of seam welds. *Why:* 5. Weld length — distortion.	sheet metal	MINOR
6	**‹bendCount› bends per part** `Fires when: inputs.bendCount !== undefined && inputs.bendCount > 8` Many bends multiply press-brake handling, tool changes and cumulative angle tolerance. **Action:** Reduce bend count, group same-tool bends, or move to a stamping die if volume supports it. *Why:* 6. High bend count — handling / setup.	sheet metal	MINOR
7	**Blank nesting ‹inputsmaterialUtilization100toFixed0›% — high skeleton scrap** `Fires when: inputs.materialUtilization !== undefined && inputs.materialUtilization < 0.65` Over a third of the sheet becomes offcut/skeleton scrap recovered only at scrap value. **Action:** Optimise nesting (common-line cutting, part rotation), right-size the blank, or negotiate scrap buy-back. *Why:* 7. Nesting utilisation.	sheet metal	MAJOR

Thermoforming — `analyseThermoformingDFM()` · 10 checks

#	Rule / trigger condition	Applies to	Sev.
1	**Draw ratio ‹drawRatio›:1 exceeds ‹drawLimit›:1 for ‹method›‹plugAssistplug›** `Fires when: !thinning.withinLimit` Beyond the draw limit the sheet thins excessively at corners, webs between features and may not form fully. *Why:* 1. Draw ratio vs the method's forming limit → webbing / excessive thinning.	thermoforming	CRITICAL
2	**Predicted min wall ‹thinningminWallMmtoFixed2› mm < required ‹need› mm** `Fires when: thinning.minWallMm < need` Corner/base walls thin below the functional minimum, risking pinholes, weak walls and poor rigidity. **Action:** Start from a thicker gauge, add plug assist for even distribution, or reduce local draw depth. *Why:* 2. Predicted thinnest wall below the functional minimum.	thermoforming	MAJOR
3	**High sag risk (index ‹sagIndex›) for ‹famtoUpperCase› at ‹unsupportedSpanMm› mm span** `Fires when: sag.risk === 'high'` The hot sheet sags under its own weight before forming, giving uneven wall thickness and possible tool contact. *Why:* 3. Sag on large unsupported spans.	thermoforming	MAJOR
4	**Moderate sag (index ‹sagIndex›) at ‹unsupportedSpanMm› mm span** `Fires when: sag.risk === 'moderate'` Some sag likely; wall distribution may drift across the sheet. *Why:* 3. Sag on large unsupported spans.	thermoforming	MINOR
5	**Internal radius ‹minInternalRadiusMm› mm < sheet thickness ‹sheetThicknessMm› mm** `Fires when: inp.minInternalRadiusMm !== undefined && inp.sheetThicknessMm && inp.minInternalRadiusMm < inp.sheetThicknessMm` Thermoforming cannot fill radii tighter than roughly the sheet gauge; corners thin severely and lose definition. **Action:** Open internal radii to ≥ 1× (ideally 2×) sheet thickness. *Why:* 4. Internal radius — sharp corners thin worst and won't form crisply.	thermoforming	MAJOR

#	Rule / trigger condition	Applies to	Sev.
6	**Draft ‹draftAngleDeg›° below ~‹draftFloor›° for thermoforming** `Fires when: inp.draftAngleDeg !== undefined && inp.draftAngleDeg < draftFloor` As the part cools it grips the tool; insufficient draft causes scuffing, distortion and ejection problems. **Action:** Increase draft to ≥ ‹draftFloor›° on drawn walls (texture needs +1°/0.025 mm depth). *Why:* 5. Draft angle — thermoformed parts need generous draft to release off the tool.	thermoforming	**MAJOR**
7	**Undercut present — needs moving core or secondary operation** `Fires when: inp.hasUndercut` Thermoform tools are largely fixed; undercuts require moving sections, stripper rings or post-form snap steps, raising tool and cycle cost. **Action:** Design the undercut out, relocate it to the trim line, or budget a moving-core tool + slower cycle. *Why:* 6. Undercuts — need moving cores / stripper plates, or must be trimmed/snapped.	thermoforming	**MAJOR**
8	**Texture over a ‹drawRatio›:1 draw may wash out** `Fires when: inp.textured && thinning && thinning.drawRatio > 1.5` On deeply drawn, thinned walls the tool texture stretches and loses depth, giving an inconsistent finish. **Action:** Apply a deeper master texture on drawn zones, or accept reduced texture depth on side walls. *Why:* 7. Texture over a deep draw → texture washes out on thinned walls.	thermoforming	**MINOR**
9	**Tolerance ±‹toleranceMm› mm tighter than thermoformable ±‹achievabletoFixed2› mm** `Fires when: inp.toleranceMm < achievable` Formed dimensions drift with sheet temperature, shrinkage and trimming; sub-process tolerances drive scrap and gauging. **Action:** Relax the tolerance, control the trim datum, or post-machine the critical feature. *Why:* 8. Tolerance realism — thermoforming is a loose-tolerance process (~±0.5% of dimension).	thermoforming	**MAJOR**
10	**‹famtoUpperCase› has a narrow forming window** `Fires when: (fam === 'pp' || fam === 'pe' || fam === 'tpo') && (inp.depthMm ?? 0) > 0` Low melt strength gives a narrow temperature window and heavy sag; forming is less forgiving than styrenics. **Action:** Use tight oven-zone control and plug assist; consider high-melt-strength (HMS) PP or a styrenic where finish allows. *Why:* 9. Formability advisory for poor-melt-strength families.	thermoforming	**OPPORTUNITY**

4 — Idea generation (37 levers)

11 cost-structure levers raised alongside the DFM findings, plus a 26-lever catalogue covering eight categories. Each carries an expected saving, a risk rating, a timeframe and a technical justification. Savings shown as a formula are computed from the part's own cost structure rather than being a fixed number.

4A Cost-structure levers

#	Rule / trigger condition	Applies to	Sev.	Saving
1	**Automate High-Labour Operations** `Fires when: labPct > 30` Labour represents ‹labPcttoFixed1›% of total part cost. Automation of repetitive tasks can reduce this significantly. *Why:* Labour automation	All commodities	process	Math.min(20, labPct * 0.4)
2	**Improve Material Utilisation via Near-Net-Shape** `Fires when: matUtil < 0.75` Current utilisation at ‹matUtil100toFixed0›%. Near-net-shape process or improved nesting targets 80–90%. *Why:* Material utilisation improvement	All commodities	material	Math.min(15, (0.85 - matUtil) * 50)
3	**OEE Improvement Programme (TPM)** `Fires when: avgOEE < 0.82` OEE at ‹avgOEE100toFixed0›% vs 85% world-class target. Each 5% OEE gain reduces effective machine cost by 5%. *Why:* OEE improvement	All commodities	process	Math.min(12, (0.85 - avgOEE) * 30)
4	**(untitled)** `Fires when: toolPct > 12` Tooling at ‹toolPcttoFixed1›% of part cost is volume-sensitive. Doubling volume halves per-part tooling cost. *Why:* Tooling cost reduction — a volume lever on an ASSUMED volume is a sensitivity note, not an instruction (a real bumper report said "increase volume" when no volume had ever been entered).	All commodities	tooling	Math.min(10, toolPct * 0.4)
5	**Overhead Rate Negotiation and Benchmarking** `Fires when: oheadPct > 15` Overhead at ‹oheadPcttoFixed1›% of part cost. Open-book costing can expose inflated factory burden. *Why:* Overhead challenge	All commodities	commercial	Math.min(6, (oheadPct - 12) * 0.4)
6	**Competitive RFQ to Reduce Supplier Margin** `Fires when: mgnPct > 12` Supplier margin at ‹mgnPcttoFixed1›% is above the 10–12% competitive benchmark. Multi-source RFQ can recover 2–5%. *Why:* Supplier margin negotiation	All commodities	commercial	Math.min(5, mgnPct - 10)
7	**Multi-Axis Machining to Consolidate Operations** `Fires when: delta > Math.max(0.05, given.costPerPart * 0.02)` Consolidating the milled and drilled content into a single 5-axis setup saves £‹deltatoFixed2›/part (‹pcttoFixed0›% of the costed machining spend of £‹givencostPerParttoFixed2›). *Why:* Claim only what the arithmetic supports: >2% of the machining spend and more than pennies.	All commodities	process	pct
8	**Regional Sourcing Study — Low-Cost Country Manufacturing** `Fires when: convPct > 30 && !alreadyLowCost` Conversion cost (process + labour) at ‹convPcttoFixed1›% creates significant regional arbitrage opportunity. *Why:* Regional sourcing — pointless advice when the part is already costed in a low-cost region (the §11 twin of this rule is gated the same way).	All commodities	commercial	Math.min(18, convPct * 0.35)
9	**(untitled)** *Why:* logistics / commercial / quality / sustainability (idea-levers.ts)	All commodities		
10	**Design Review for DFM Compliance** `Fires when: costOptimisations.length < 5` Formal DFM review with manufacturing engineering to identify part features that increase cost without functional benefit. *Why:* Ensure we have 5–8 items — add generic ones if needed	All commodities	design	5%
11	**Annual Volume Re-Commitment for Better Pricing** `Fires when: costOptimisations.length < 6 && !volumeAssumed` Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	All commodities	commercial	3%

4B 360-degree lever catalogue — 8 categories

Commercial · 4 levers

#	Lever	Applies to	Owner	Saving
6	**Raw-Material Price Indexation Clause** `Offered when: matPct > 40` With material at ‹matPcttoFixed1›% of cost, the quote embeds the supplier's hedge against metal/resin volatility. An index-linked clause removes that risk premium from the piece price. *Basis:* Indexing the material content to LME/CRU/resin indices with quarterly true-up removes the 2–5% volatility contingency suppliers price in, and makes future raw-material moves transparent in both directions.	All commodities	sourcing / Low / Quick Win	`Math.min(3, matPct * 0.05)`
24	**Payment Terms / Early-Settlement Discount** `Offered when: mgnPct > 10` A supplier pricing ‹mgnPcttoFixed1›% margin values cash. Early settlement (e.g. 14 days vs 60) is routinely worth 1–2% off the piece price and costs the buyer only working capital. *Basis:* Standard dynamic-discounting maths: 2%/46 days ≈ 16% annualised — attractive to any supplier financing at SME rates. A pure win where the buyer's cost of capital is lower than the supplier's.	All commodities	sourcing / Low / Quick Win	`2`
25	**Make-vs-Buy Assessment** `Offered when: oheadPct > 18 && (procPct + labPct) > 35` Overhead at ‹oheadPcttoFixed1›% on top of ‹procPctlabPcttoFixed1›% conversion means a large share of this price is another factory's burden. If volumes justify a cell, in-housing removes overhead and margin together. *Basis:* Make-vs-buy on this should-cost basis: in-house cost = material + conversion at own rates + own burden. Parts with high supplier burden and mature process technology are the standard insourcing candidates.	All commodities	sourcing / High / Long Term	`Math.min(6, oheadPct * 0.2)`
26	**Learning-Curve Price-Down Agreement** `Offered when: !input.learningCurve?.enabled && labPct > 20 && ctx?.volumeProvided === true` Labour is ‹labPcttoFixed1›% of cost and no learning curve is priced in. On a multi-year programme, labour productivity improves whether or not the contract captures it — the contract should. *Basis:* Wright's-law experience curves (85–95% typical in assembly-heavy work) imply 5–15% labour-cost reduction per volume doubling. An agreed annual price-down of 2–3% simply shares an improvement that will happen anyway.	All commodities	sourcing / Low / Quick Win	`3`

Design · 3 levers

#	Lever	Applies to	Owner	Saving
7	**Tolerance and Surface-Finish Relaxation** `Offered when: tightShare > 0.03 || (['machining', 'cast_and_machine'].includes(commodity) && procPct > 35)` *Basis:* Industry DFM studies attribute 20–40% of precision-machining cost to the tightest decile of callouts. Opening a ±0.01 to ±0.05 where function allows can delete a grinding operation outright.	machining, cast and machine	design / Low / Quick Win	`tightShare > 0.03 ? Math.min(6, tightShare * 100 * 0.8) : 4`
8	**Part-Count / Feature Integration Review (Boothroyd)** `Offered when: opCount >= 5 && ['machining', 'cast_and_machine', 'sheet_metal', 'sheet_metal_fab', 'biw_assembly', 'pcba', 'wiring_harness'].includes(commodity)` ‹opCount› operations suggest features or joined parts that may not need to exist separately. The Boothroyd minimum-part-count test asks of each: does it move, must it be a different material, must it be separate for assembly? *Basis:* Boothroyd–Dewhurst DFA practice: any part failing all three criteria is a candidate to integrate into its neighbour. Integrating a bracket into a casting or forming a feature instead of fastening it removes its entire operation chain.	machining, cast and machine, sheet metal, sheet metal fab, biw assembly, pcba, wiring harness	design / Medium / Long Term	`6`
9	**Lightweighting / Wall Optimisation** `Offered when: (commodity === 'casting' || commodity === 'rotational_moulding' || MOULD_FAMILY.includes(commodity)) && matPct > 42` Material is ‹matPcttoFixed1›% of cost and this is a wall-driven process — every 10% of wall thickness removed is roughly 10% off the material line and a shorter cooling/solidification cycle on top. *Basis:* Topology/wall studies against the measured geometry (the kernel already knows the wall map) routinely find 8–15% mass with no load-path compromise. Cycle time falls with the square of wall on cooling-limited parts.	casting, rotational moulding, MOULD FAMILY	design / Medium / Long Term	`Math.min(6, 3 + (matPct - 42) * 0.15)`

Logistics · 4 levers

#	Lever	Applies to	Owner	Saving
18	**(untitled)** `Offered when: pkgPct > 4` Packaging is ‹pkgPcttoFixed1›% of part cost. *Basis:* Returnables convert a per-part consumable into an amortised asset: typical automotive experience is 40–70% packaging cost reduction on stable volume routes, plus damage-rate and waste-stream benefits.	All commodities	Low / Quick Win	`Math.min(4, pkgPct * 0.5)`
19	**Pack Density / Cube Utilisation Review** `Offered when: (pkgPct + logPct) > 8` Packaging plus logistics is ‹pkgPctlogPcttoFixed1›% of part cost. Freight is bought by the cubic metre — parts that nest, stack or ship in denser dunnage cut it directly. *Basis:* Nesting orientation studies and custom dunnage regularly lift parts-per-container 20–50%. On intercontinental routes this is often worth more than any process improvement on the part.	All commodities	Low / Quick Win	`Math.min(4, (pkgPct + logPct) * 0.3)`
20	**Freight Mode and Consolidation Review** `Offered when: logPct > 6 && !alreadyLowCost` Logistics is ‹logPcttoFixed1›% of part cost. Mode (sea vs air), consolidation (milk runs, full containers) and incoterm are all buyer-controlled levers. *Basis:* Air-to-sea conversion cuts freight 70–85% where lead time allows; consolidated FCL versus LCL saves 20–40%. The landed-cost module prices these scenarios per country on request.	All commodities	Low / Quick Win	`Math.min(5, logPct * 0.5)`
21	**Total Landed Cost — Near-Shore Check** `Offered when: alreadyLowCost && logPct > 9` This part is costed in a low-cost region but logistics is ‹logPcttoFixed1›% of its cost. At that freight share, a nearer supplier at a higher piece price can win on landed cost — the 20-country comparison quantifies it. *Basis:* Freight, duty, inventory-in-transit and expedite risk offset ex-works savings on heavy or bulky parts. Run the regional comparison on landed (not ex-works) cost before renewing the source.	All commodities	sourcing / Medium / Long Term	`Math.min(6, logPct * 0.5)`

Material · 4 levers

#	Lever	Applies to	Owner	Saving
1	**Alternate Material Grade Study** `Offered when: METAL_FORM.includes(commodity) && matPct > 40` Material is ‹matPcttoFixed1›% of part cost — the largest lever on this part is the metal itself. An equivalent lower-cost grade (or secondary/remelt alloy where the spec allows) attacks it directly. *Basis:* Grade substitution within the same family (e.g. 6082→6060 where strength allows, primary→secondary casting alloy, DP600→HSLA where CAE confirms) typically saves 5–15% of the material line with no process change. Requires engineering sign-off against the load case.	METAL FORM	design / Medium / Medium Term	`Math.min(8, 3 + (matPct - 40) * 0.15)`
2	**Closed-Loop Regrind of Runners and Trim** `Offered when: MOULD_FAMILY.includes(commodity) && matUtil < 0.92` ‹1matUtil100toFixed0›% of the resin bought does not ship in the part. Reprocessing runners, sprues and trim as a controlled regrind blend recovers most of that value. *Basis:* A 10–20% regrind blend is standard practice for non-optical, non-safety mouldings and is invisible in the part when dried and dosed correctly. The saving is the resin price times the recovered fraction.	MOULD FAMILY	supplier / Low / Quick Win	`Math.min(5, (0.95 - matUtil) * 40)`
3	**Scrap Revenue Recovery at Index Prices** `Offered when: ['machining', 'cast_and_machine', 'sheet_metal', 'sheet_metal_fab', 'forging'].includes(commodity) && matUtil < 0.80` ‹1matUtil100toFixed0›% of the bought material leaves as chips, offcuts or skeleton. That scrap has an index price — the credit belongs in the piece price, not in the supplier's margin. *Basis:* Clean segregated scrap (Al swarf, steel skeleton, electrical-steel stamping web) trades at 20–60% of prime price. An open-book scrap credit clause at LME/index-linked prices is a standard automotive term.	machining, cast and machine, sheet metal, sheet metal fab, forging	sourcing / Low / Quick Win	`Math.min(4, 1 + (0.80 - matUtil) * 15)`

#	Lever	Applies to	Owner	Saving
5	**Consumables Rationalisation (Cores, Patterns, Shell)** `Offered when: consumShare > 0.20` Per-part consumables are ‹consumShare100toFixed0›% of the material line. Core count, pattern life and shell recipe are all design-controllable. *Basis:* Core consolidation (one complex core replacing two simple ones), longer-life wax/pattern tooling, and shell-sand reclaim each cut the recurring consumable line 10–30% in foundry practice.	All commodities	design / Medium / Medium Term	`Math.min(6, consumShare * 20)`

Process · 4 levers

#	Lever	Applies to	Owner	Saving
10	**Attack the Bottleneck: "‹name›"** `Offered when: topOp && topOpShare > 0.5 && opCount >= 2 && convTotal / tot > 0.2` One operation — ‹name› — carries ‹topOpShare100toFixed0›% of the conversion cost. Improving anything else first is working on the wrong problem. *Basis:* Focused cycle-time engineering on the governing operation (feeds/speeds, tooling upgrade, cooling optimisation, SMED on its changeover) moves the whole part cost; a 20% gain on the bottleneck is a realistic first-pass target.	All commodities	supplier / Medium / Medium Term	`Math.min(8, topOpShare * (procPct + labPct) * 0.2)`
11	**(untitled)** `Offered when: CAVITY_PROCESSES.includes(commodity) && ops.length > 0 && ops.every(o => (o.partsPerCycle ?? 1) <= 1) && (procPct + labPct) > 25` Every cycle currently makes one part. A 2-cavity (or family) tool halves the machine and labour minutes per part against roughly 60–80% more tooling NRE. *Basis:* Cavitation is the standard volume lever in moulding and die casting: machine time per part scales with 1/cavities while the tool price scales far slower (~1.6–1.8× for 2 cavities). The tool already cost-ranks cavity options in the derivation trace.	CAVITY PROCESSES	supplier / Medium / Medium Term	`Math.min(12, (procPct + labPct) * 0.3)`
12	**Unmanned / Lights-Out Running** `Offered when: ['machining', 'cast_and_machine'].includes(commodity) && labPct > 12 && avgOEE >= 0.75` Labour is ‹labPcttoFixed1›% of cost on a machine-paced part with healthy OEE (‹avgOEE100toFixed0›%) — the classic profile for bar-feed or pallet-pool unattended running. *Basis:* Bar feeders, pallet pools and probing let a stable machining process run a third shift with no operator. Suppliers typically pass through 30–50% of the attended-labour saving on committed volumes.	machining, cast and machine	supplier / Medium / Medium Term	`Math.min(6, labPct * 0.3)`
13	**Multi-Machine Manning** `Offered when: maxManning >= 1 && labPct > 15 && !['wiring_harness', 'composites', 'biw_assembly'].includes(commodity) && ops.some(o => (o.manning ?? 0) >= 1 && o.cycleTimeHr > (o.labourTimeHr ?? 0) * 0.8)` Operations are manned at up to ‹maxManningtoFixed1› operators per machine while the machine paces the cycle. One operator tending two machines is the textbook correction. *Basis:* Where the operator loads, starts and walks away, 1:2 or 1:3 manning halves or thirds the charged labour minutes. Requires cell layout and staggered cycles — a supplier kaizen, not an investment.	wiring harness, composites, biw assembly	supplier / Low / Quick Win	`Math.min(5, labPct * 0.25)`

Quality · 2 levers

#	Lever	Applies to	Owner	Saving
22	**Right-Size Inspection and Test** `Offered when: hasOp(RX_INSPECT) && inspShare > 0.05` Inspection/test operations carry ‹inspShare100toFixed1›% of part cost as separate steps. Capable processes earn reduced inspection — that is what capability data is for. *Basis:* SPC-based skip-lot inspection, in-line gauging on the machine, and poka-yoke at the source typically cut standalone inspection cost 40–70% once Cpk ≥ 1.33 is demonstrated — without raising escape risk.	All commodities	supplier / Low / Medium Term	`Math.min(4, inspShare * 100 * 0.6)`

#	Lever	Applies to	Owner	Saving
23	**Eliminate Manual Finishing at Source** `Offered when: hasOp(RX_REWORK) && reworkShare > 0.02` Deburring/fettling/rework operations carry ‹reworkShare100toFixed1›% of part cost. Manual finishing is a symptom — the cause is upstream (tool wear, gating, die condition) and cheaper to fix there. *Basis:* Sharp tooling discipline, gate/overflow redesign and die maintenance remove the burr instead of removing it by hand. Where finishing must remain, tumbling/vibratory processes beat manual labour 3:1.	All commodities	supplier / Low / Medium Term	`Math.min(4, 1 + reworkShare * 100 * 0.5)`

Sustainability · 2 levers

#	Lever	Applies to	Owner	Saving
4	**Recycled-Content (PCR) Resin Blend** `Offered when: MOULD_FAMILY.includes(commodity) && matPct > 35` Resin is ‹matPcttoFixed1›% of the part. A 10–30% post-consumer-recycled blend cuts both cost and the part's carbon footprint where colour and mechanicals allow. *Basis:* PCR PP/ABS trades 15–30% below prime. Non-appearance and black parts are the natural first candidates; many OEM sustainability targets now require declared recycled content anyway.	MOULD FAMILY	design / Medium / Medium Term	`Math.min(4, matPct * 0.08)`
14	**Energy Productivity Programme** `Offered when: ENERGY_INTENSE.includes(commodity) || hasOp(RX_HEAT)` This process is energy-intensive (melt, cure, oven or heat-treat content). Energy sits inside the machine rate and overhead — and it is the fastest-moving input cost of the decade. *Basis:* Melt/furnace scheduling to off-peak tariffs, regenerative burners, cure-recipe optimisation and compressed-air leak programmes typically cut process energy 10–20%, of which 2–4% of part cost reaches the piece price — and the carbon line falls with it.	ENERGY INTENSE	supplier / Low / Medium Term	`3`

Tooling · 3 levers

#	Lever	Applies to	Owner	Saving
15	**Tooling Ownership and End-of-Life Terms** `Offered when: toolingNRE > 0 && toolPct > 10` £‹toolingNREtoLocaleString› of tooling sits behind this part. Who owns it, who maintains it, and what happens at end of programme are worth points of piece price — and all of it is negotiable before the tool is cut. *Basis:* A separate tooling PO with customer ownership prevents the tool being re-amortised inside future piece prices, secures the asset for resourcing leverage, and makes tool maintenance an explicit (auditable) line instead of a hidden uplift.	All commodities	sourcing / Low / Quick Win	`Math.min(4, toolPct * 0.2)`
16	**Soft / Bridge Tooling for This Volume** `Offered when: amortVol > 0 && amortVol < 25000 && toolPct > 25` Tooling is ‹toolPcttoFixed1›% of part cost over only ‹amortVoltoLocaleString› parts. Below ~25k parts, aluminium tools, printed inserts or master-frame systems beat hard steel tooling. *Basis:* Aluminium production tools cost 40–60% of P20 steel and comfortably survive 10–100k shots in polyolefins; modular master frames share the base across a family. The right answer flips back to steel only when volume grows.	All commodities	supplier / Medium / Medium Term	`Math.min(8, toolPct * 0.3)`
17	**Tool-Life Extension Programme** `Offered when: ['casting', 'forging', 'sheet_metal', 'sheet_metal_fab'].includes(commodity) && toolPct > 15` Tooling at ‹toolPcttoFixed1›% of part cost means every extra thousand shots of die life is money. Surface treatment and disciplined maintenance are the cheapest capacity there is. *Basis:* Nitriding/PVD on wear surfaces, weld-repair schedules and hot-die lubrication programmes extend die life 30–100% in HPDC and forging practice — directly diluting the NRE per part.	casting, forging, sheet metal, sheet metal fab	supplier / Low / Medium Term	`Math.min(4, toolPct * 0.15)`

Appendix — Provenance and how to regenerate

This document is generated, not written. Two commands rebuild it against whatever the engine currently contains:

```
npx tsx scripts/extract-rule-library.ts
python3 scripts/build-rule-library-pdf.py
```

The extractor parses the source directly: it walks each rule's enclosing scope to find the commodity gate in force, captures the guard condition verbatim, and reads severity, category, saving, risk and recommendation out of the object literal the engine constructs. Line numbers below are where each rule lives, so any entry here can be traced to source in one step.

File	Contents
src/engine/dfm-dfa.ts	41 DFM rules, 11 DFA rules, 11 cost-structure levers
src/engine/idea-levers.ts	26-lever 360-degree catalogue
src/engine/modules/*-advisor.ts	10 geometry advisors, 75 process-physics checks
src/engine/opportunity-ranking.ts	Ranks every finding by saving; root-sum-square headline capped at 40%
src/engine/should-cost-audit.ts	Pre-flight lesson checks that block an implausible estimate before it is published

One limitation worth stating. The trigger conditions are reproduced exactly, but a rule whose threshold depends on the commodity benchmark table (utilisation, OEE, bucket percentages) will read as a comparison against a variable. The benchmark values themselves live in `src/engine/insights.ts` and are not reproduced here.